SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 2 – Scenario-Based Questions (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 2: Scenario-Based Questions
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	2024-SLOT A	**Date:**	28-10-2026

Code of Conduct

I S.SRIJITHA 192471024 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: srijitha

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Scenario-Based Qns

Q1. Image Enhancement Using Grey-Level Transformations

A hospital uses a digital imaging system to capture X-ray images, but some images appear too dark to identify important anatomical structures. The image processing team decides to apply grey-level transformation techniques to improve image visibility. Apply a suitable transformation to enhance the brightness and contrast of the image. Calculate the transformed intensity values for selected pixels using the chosen transformation. Explain how the transformation improves the visibility of important features in the X-ray image.

Q2. Histogram Processing and Spatial Filtering

A security surveillance system captures images at night, resulting in poor contrast and unwanted noise. The system must enhance the images to identify people and objects more clearly. Apply histogram equalization to improve the contrast of the captured image. Use an appropriate spatial filtering technique to reduce noise while preserving important details. Compare the effects of both techniques and explain how they improve the quality of the surveillance images.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

A satellite imaging system captures land images containing noise and blurred boundaries between different geographical regions. The image processing team decides to use frequency domain filtering to improve image quality. Apply a low-pass filter to reduce high-frequency noise and smooth the image. Use a high-pass filter to enhance edges and sharpen important geographical features. Explain the differences between the two filters and determine how each technique contributes to image interpretation.

Q4. Image Segmentation Using Edge Detection and Thresholding

An automated manufacturing system must inspect metal components to identify cracks and surface defects. The captured images contain variations in brightness, making defect detection difficult. Apply an appropriate edge detection technique to identify the boundaries of possible cracks. Use thresholding to separate defective regions from the background and construct a binary image representing the detected defects. Explain how the selected techniques help the inspection system identify defective components.

Q5. Region-Based Segmentation and Boundary Detection

An agricultural monitoring system uses aerial images to identify crop fields, water bodies, and surrounding land regions. The captured images contain regions with similar intensity values and some discontinuous boundaries. Apply region-based segmentation to group neighbouring pixels with similar characteristics. Use edge linking and boundary detection techniques to connect broken edges and identify the boundaries of different regions. Explain how these techniques help distinguish agricultural fields and improve the accuracy of land-region identification.

Annexure A – Scenario-Based Questions

Q1. Image Enhancement Using Grey-Level Transformations

Problem Statement

X-ray images captured by the hospital imaging system are too dark, so important anatomical structures cannot be identified. A grey-level (point) transformation must be applied to improve brightness and contrast, the transformed values must be calculated for selected pixels, and the improvement in visibility must be explained.

1. Choice of Transformation

A grey-level transformation maps every input intensity *r* to an output intensity *s* using s = T(r), independently of neighbouring pixels. For a dark image, most pixels occupy a narrow band of low intensities, so the transformation should stretch the dark range and compress the bright range. The **power-law (gamma) transformation** is selected:

$$s = (L - 1) \cdot (r / (L - 1))^{\gamma}$$

where L = 256 for an 8-bit image and γ is the gamma value. For γ < 1 the curve lies above the straight line s = r, so dark pixels are mapped to much brighter values (as in the curves of Figure 1). Here **γ = 0.5** is chosen, giving $s = 255 \cdot (r / 255)^{0.5}$. Gamma is preferred over the log transformation s = c·log(1 + r) because its curvature can be tuned through γ, whereas the log curve is fixed and brightens the image more aggressively.

2. Calculation of Transformed Intensities

Worked example for r = 60: r / 255 = 0.2353; $(0.2353)^{0.5} = 0.4851$; s = 255 × 0.4851 = 123.69 ≈ **124**. The same steps give the values for the other selected pixels below.

Table 1: Gamma transformation (γ = 0.5, L = 256) for selected pixels

Input r	r / 255	$(r / 255)^{0.5}$	$s = 255 \times (r / 255)^{0.5}$	Output s (rounded)	Gain (s − r)
20	0.0784	0.2801	71.41	71	51
40	0.1569	0.3961	101.00	101	61
60	0.2353	0.4851	123.69	124	64
80	0.3137	0.5601	142.83	143	63
100	0.3922	0.6262	159.69	160	60
150	0.5882	0.7670	195.58	196	46
200	0.7843	0.8856	225.83	226	26
255	1.0000	1.0000	255.00	255	0

3. Result

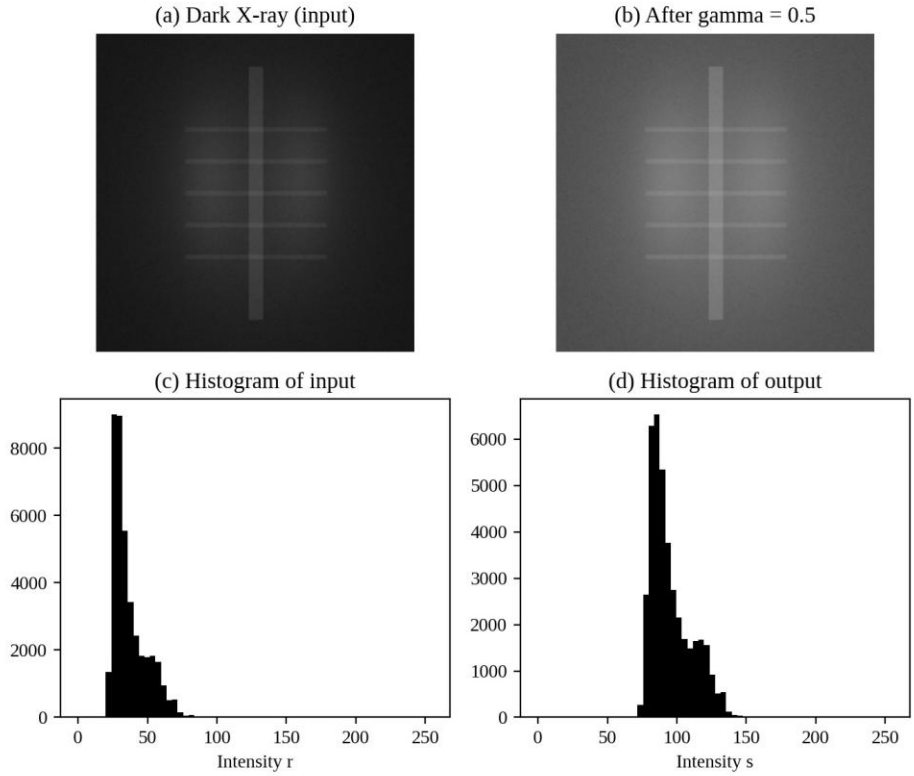


Figure 1(a): Dark X-ray image before and after gamma correction, with histograms

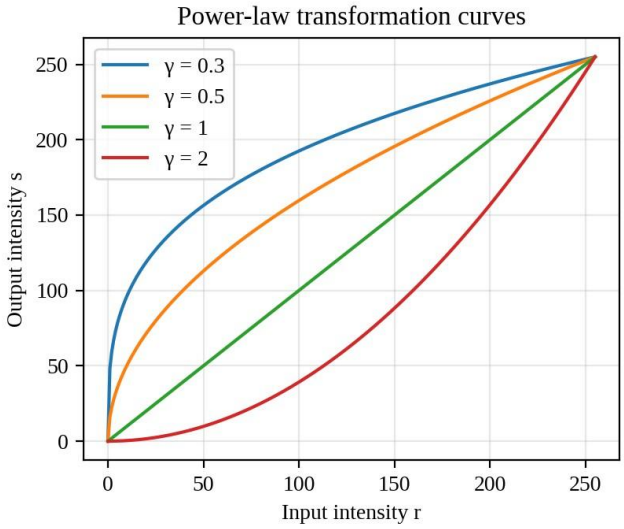


Figure 1(b): Power-law curves for different gamma values

4. How the Transformation Improves Visibility

- **Expansion of dark levels:** inputs between 20 and 40 (a difference of 20 grey levels) become 71 and 101, a difference of about 30 levels. Tissue and bone that differ only slightly in the dark range therefore become easier to separate.
- **Compression of bright levels:** inputs between 200 and 255 (55 levels apart) become 226 and 255 (29 levels apart). This prevents already-bright regions from saturating.

- **Histogram shift:** as seen in Figure 1(a), the histogram moves from the extreme left towards the centre, increasing both the mean brightness and the spread (contrast) of the image.
- **Limitation:** if γ is too small, the image looks washed out and noise in dark regions is also amplified. The value of γ is therefore tuned (typically 0.4 – 0.6 for X-rays) and checked visually by the radiologist.

Q2. Histogram Processing and Spatial Filtering

Problem Statement

Night-time surveillance images have poor contrast and unwanted noise. Histogram equalization is applied to improve contrast, a spatial filter is used to reduce noise while preserving details, and the two techniques are compared.

1. Histogram Equalization

Histogram equalization (HE) redistributes intensity values so that the output histogram is approximately uniform. For an image with N pixels and L grey levels, the transformation is the scaled cumulative distribution function (CDF):

$$s_k = (L - 1) \cdot \Sigma_{j=0}^{k} n_j / N$$

Worked example: an 8 × 8 night image (N = 64) with 3-bit grey levels (L = 8). Most pixels lie in the dark levels 0 – 2.

Table 2: Histogram equalization of an 8 × 8, 3-bit image

Level r_k	n_k	$p(r_k) = n_k / 64$	CDF	$s = 7 \times$ CDF	Rounded s_k
0	15	0.2344	0.2344	1.641	2
1	18	0.2813	0.5156	3.609	4
2	12	0.1875	0.7031	4.922	5
3	8	0.1250	0.8281	5.797	6
4	5	0.0781	0.9063	6.344	6
5	3	0.0469	0.9531	6.672	7
6	2	0.0313	0.9844	6.891	7
7	1	0.0156	1.0000	7.000	7

The mapping is 0→2, 1→4, 2→5, 3→6, 4→6, 5→7, 6→7, 7→7. The new histogram becomes: level 2 → 15 pixels, level 4 → 18, level 5 → 12, level 6 → 8 + 5 = 13, level 7 → 3 + 2 + 1 = 6 (total 64). Before equalization 45 of 64 pixels (70%) occupied levels 0 – 2; afterwards the pixels spread over levels 2 – 7, so the dynamic range of the image is used fully and contrast improves.

2. Spatial Filtering for Noise Reduction

Night images captured at high ISO contain impulse (salt-and-pepper) noise and Gaussian noise. A **3 × 3 median filter** is chosen because it replaces each pixel by the median of its neighbourhood, removing isolated extreme values without blurring edges. Example for a neighbourhood whose centre pixel (255) is a noise spike:

Table 3: 3 × 3 neighbourhood containing an impulse

12	14	13
15	255	16
14	13	15

Sorted values: 12, 13, 13, 14, 14, 15, 15, 16, 255. The median is the 5th value = **14**, so the spike is replaced by 14. A 3 × 3 mean (averaging) filter would give (12 + 14 + 13 + 15 + 255 + 16 + 14 + 13 + 15) / 9 = 40.8, which still leaves a bright smear and would blur genuine edges. The median filter is therefore the better choice for preserving the outlines of people and objects.

3. Results and Comparison

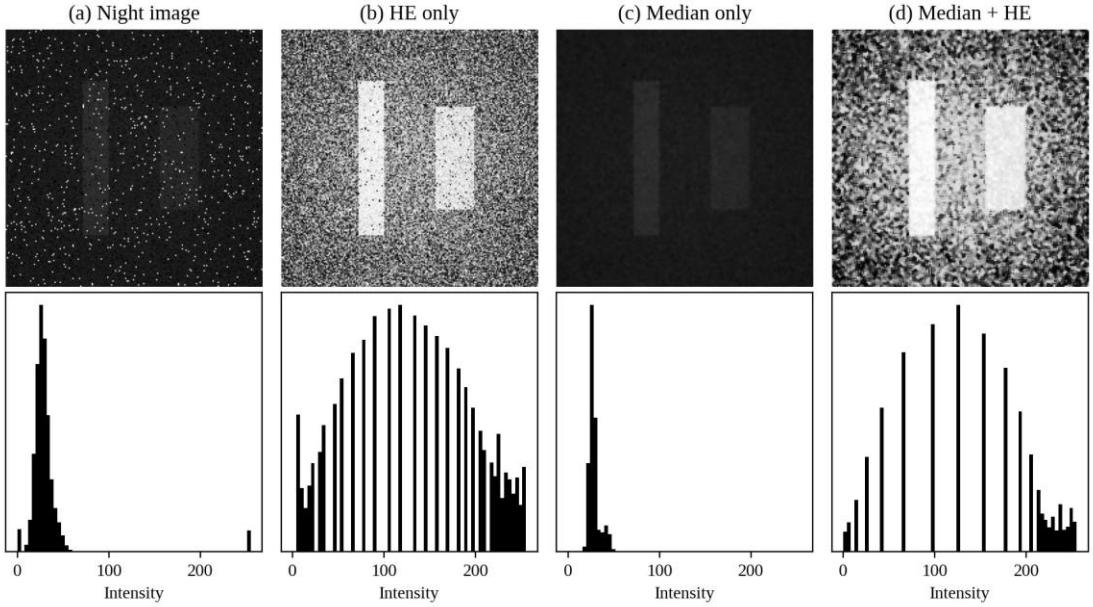


Figure 2: (a) night image, (b) HE only, (c) median filter only, (d) median filter followed by HE

Figure 2(b) shows that equalization alone brightens the scene but also makes the noise much more visible, because HE stretches noise along with the signal. Figure 2(c) removes the noise but the image remains dark. Figure 2(d), where the median filter is applied first and HE second, gives the best result: clean and high-contrast. In practice, noise removal is therefore performed before equalization.

Table 4: Comparison of histogram equalization and median filtering

Aspect	Histogram Equalization	Median Spatial Filter
Type of operation	Point operation using the global histogram	Neighbourhood operation using a local window
Main purpose	Improve contrast	Reduce noise
Effect on noise	Amplifies noise	Suppresses impulse noise
Effect on edges and details	Makes details more visible by stretching intensities	Preserves edges, removes isolated spikes
Effect on histogram	Flattens and spreads the histogram	Histogram shape largely unchanged
Limitation	May over-enhance background	Window too large removes fine details

Conclusion: median filtering followed by histogram equalization gives surveillance images in which people and objects are clearly separated from the background, which improves identification.

Q3. Image Smoothing and Sharpening Using Frequency Domain Filtering

Problem Statement

Satellite images contain noise and blurred boundaries between regions. A low-pass filter is used to reduce noise (smoothing) and a high-pass filter is used to enhance edges (sharpening). The differences between the filters and their contribution to image interpretation must be explained.

1. Frequency-Domain Filtering Procedure

- Pad the M × N image f(x, y) to size P × Q = 2M × 2N to avoid wrap-around error.
- Multiply by $(-1)^{x+y}$ to move the origin of the spectrum to the centre.
- Compute the Discrete Fourier Transform F(u, v).
- Multiply by the filter transfer function: G(u, v) = H(u, v) · F(u, v).
- Compute the inverse DFT, take the real part, multiply again by $(-1)^{x+y}$, and crop to M × N.

2. Low-Pass and High-Pass Filters

Noise and fine texture lie in the high frequencies, while smooth regions lie in the low frequencies. With D(u, v) the distance from the centre of the spectrum and D_0 the cut-off frequency, the Gaussian filters are used, because the Gaussian has no ringing (unlike the ideal filter):

$$H_{LP}(u, v) = \exp[- D^2(u, v) / (2 D_0^2)] \qquad H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

Numerical example with $D_0 = 30$:

Table 5: Gaussian filter response for $D_0 = 30$

D(u, v)	$H_{LP} = \exp(-D^2 / 1800)$	$H_{HP} = 1 - H_{LP}$	Interpretation
0	1.000	0.000	DC component (average brightness) kept by LPF, removed by HPF
10	0.946	0.054	Low frequency: smooth regions
20	0.801	0.199	Low frequency: smooth regions
40	0.411	0.589	Transition band
60	0.135	0.865	Edges / fine detail
90	0.011	0.989	Mostly noise

A component at D = 90 (mostly noise) passes through the low-pass filter with only 1.1% of its amplitude, while a component at D = 10 passes with 94.6%. The high-pass filter does the opposite: components at D = 60 pass with 86.5% of their amplitude, while the DC term is removed.

3. Results

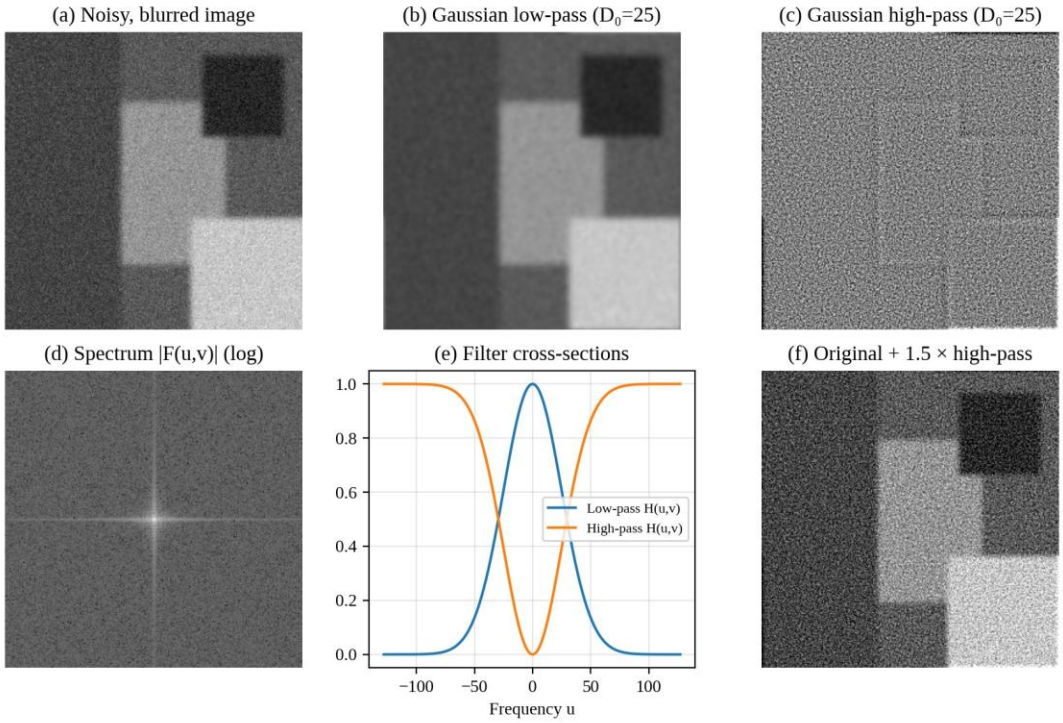


Figure 3: Low-pass smoothing and high-pass sharpening of a noisy satellite image

Figure 3(b) shows that noise is strongly reduced and the land regions become uniform, but boundaries are slightly blurred. Figure 3(c) shows only the edges and noise (the DC term is removed, so the display is shifted to mid-grey). Adding the high-pass output back to the original, $g = f + k \cdot f_{HP}$ with $k = 1.5$, gives the sharpened image in Figure 3(f). Since a high-pass filter also amplifies noise, low-pass smoothing should be applied before sharpening.

4. Differences Between the Two Filters

Table 6: Low-pass versus high-pass filtering

Aspect	Low-Pass Filter	High-Pass Filter
Frequencies passed	Low frequencies (below D_0)	High frequencies (above D_0)
Frequencies attenuated	High frequencies (noise, edges)	Low frequencies (smooth areas, DC)
Effect on image	Smoothing / blurring	Sharpening / edge enhancement
Effect on noise	Reduces noise	Amplifies noise
Effect on boundaries	Blurs region boundaries	Highlights region boundaries
Transfer function	$H_{LP} = \exp(-D^2 / 2D_0^2)$	$H_{HP} = 1 - H_{LP}$
Use in this scenario	Noise removal, uniform land-cover areas	Coastlines, roads, field and region borders

5. Contribution to Image Interpretation

The low-pass filter makes large homogeneous regions such as water bodies, forests and farmland uniform, which makes visual interpretation and land-cover classification more reliable. The high-pass filter emphasises the boundaries between these regions, so coastlines, rivers, roads and field borders can be located accurately. Used together (smooth first, then sharpen), they give a satellite image that is both clean and sharp.

Q4. Image Segmentation Using Edge Detection and Thresholding

Problem Statement

An automated system must inspect metal components for cracks and surface defects, but brightness varies across the images. Edge detection is used to find crack boundaries, thresholding is used to separate defects from the background, and a binary defect image is constructed.

1. Processing Steps

- Smooth the image with a Gaussian filter ($\sigma \approx 1 - 1.5$) to suppress sensor noise.
- Apply the **Sobel operator** to compute the gradient and detect crack boundaries.
- Apply **thresholding** to produce a binary image in which defect pixels = 1 and background = 0.
- Remove small isolated objects by morphological opening, then label connected components and measure their area and length.

2. Edge Detection with the Sobel Operator

Sobel masks estimate the horizontal and vertical derivatives; the neighbourhood is labelled z_1 to z_9 (row by row). Gradient components and magnitude:

$$G_x = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7) \qquad G_y = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$|\nabla f| = \sqrt{(G_x^2 + G_y^2)} \qquad \theta = \tan^{-1}(G_y / G_x)$$

Worked example: a 3 × 3 patch containing the edge of a dark crack (intensity 40) next to the bright surface (intensity 100):

Table 7: 3 × 3 neighbourhood across a crack edge

100	100	40
100	100	40
100	100	40

$G_x = (40 + 2\times40 + 40) - (100 + 2\times100 + 100) = 160 - 400 = -240$; $G_y = (100 + 200 + 40) - (100 + 200 + 40) = 0$. Therefore $|\nabla f| = \sqrt{(240^2 + 0^2)} = \mathbf{240}$ and $\theta = 180°$, meaning the gradient is horizontal and the edge is vertical. With threshold T = 100, 240 > 100, so the centre pixel is marked as an **edge pixel**. In flat areas $|\nabla f| \approx 0$, so no edge is marked.

3. Thresholding to Obtain the Binary Defect Image

For intensity thresholding the **iterative global threshold** algorithm is used: (i) start with T = mean intensity; (ii) divide pixels into G_1 (> T) and G_2 (≤ T); (iii) compute their means μ_1, μ_2; (iv) set $T = (\mu_1 + \mu_2) / 2$; (v) repeat until T stops changing. Example on a 4 × 4 patch where cracks are dark:

Table 8: 4 × 4 grey-level patch

190	185	45	188
192	40	38	186
187	42	184	190
189	191	186	185

Initial T_0 = mean = 151.1. Then $\mu_1 = 187.75$ (12 pixels above T_0) and $\mu_2 = 41.25$ (4 pixels below), so $T_1 = (187.75 + 41.25) / 2 = 114.5$. Repeating the step gives the same groups and T = 114.5, so the algorithm has converged. Pixels with intensity below 114.5 are set to 1 (defect), giving the binary image:

Table 9: Binary defect image (1 = defect, 0 = background)

0	0	1	0
0	1	1	0
0	1	0	0
0	0	0	0

Because the brightness of real metal images changes across the surface, a single global threshold fails (Figure 4(d)): the dark side of the image is wrongly classified as defect. Using a **local (adaptive) threshold**, where each pixel is compared with the mean of its neighbourhood, T(x, y) = local mean − C, gives the correct crack shown in Figure 4(e).

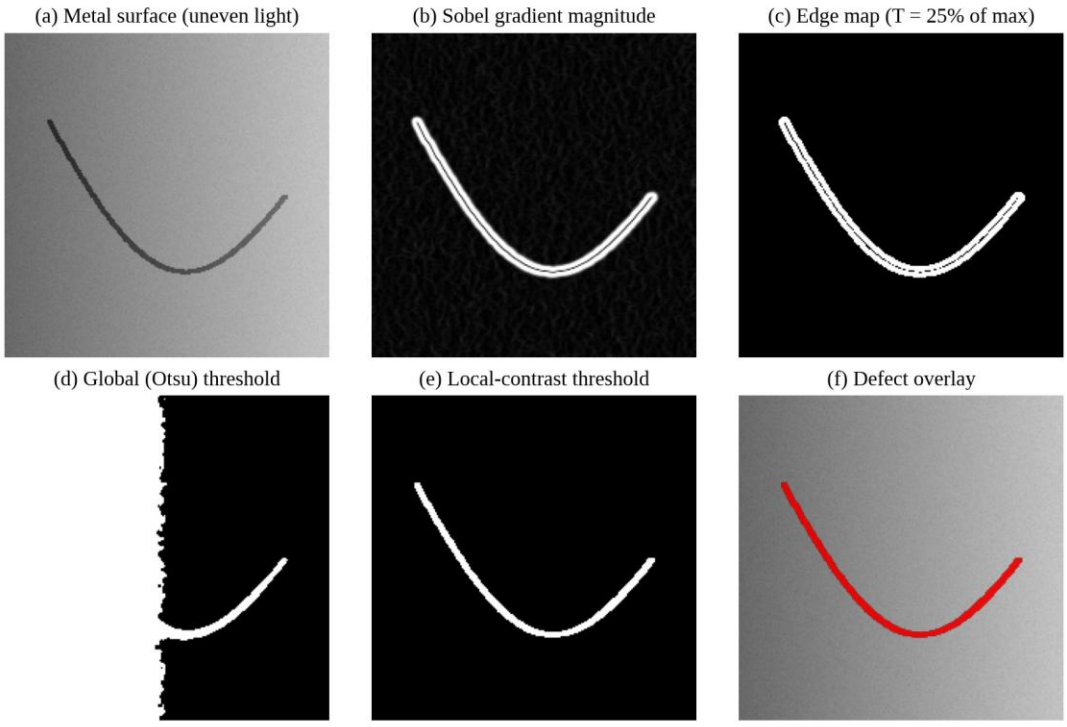


Figure 4: Sobel edge detection and thresholding on a metal surface with uneven illumination

4. How the Techniques Help the Inspection System

- **Edge detection** gives the exact boundary of the crack, from which length, shape and orientation can be measured.
- **Thresholding** isolates the whole defective region as a binary object, so area and pixel count can be measured easily.
- **Decision:** a component is rejected if the labelled defect area or length exceeds the acceptable limit, which allows fast and consistent automatic inspection.
- Adaptive thresholding makes the system robust to brightness variations, which reduces false rejects.

Q5. Region-Based Segmentation and Boundary Detection

Problem Statement

Aerial images of farmland contain crop fields, water bodies and surrounding land. Some regions have similar intensities and some boundaries are discontinuous. Region-based segmentation groups similar neighbouring pixels, and edge linking and boundary detection connect broken edges to identify region boundaries.

1. Region-Based Segmentation (Region Growing)

Region growing starts from seed pixels and adds 4-connected neighbours whose intensity satisfies the similarity condition $|f(x, y) - f(\text{seed})| \leq T$, until no more pixels can be added. It uses the fact that pixels in the same field are both similar in intensity and spatially connected.

Worked example: 5 × 5 image, T = 5, 4-connectivity, three seeds: A = (row 2, col 2) with value 12, B = (row 1, col 5) with value 55, C = (row 4, col 1) with value 48.

Table 10: Input image

12	11	13	52	55
10	12	14	54	53
11	13	50	51	56
48	12	56	52	54
47	46	50	53	55

Region A accepts values from 7 to 17, region B from 50 to 60, and region C from 43 to 53. Growing gives the label map below (1 = A, 2 = B, 3 = C):

Table 11: Segmented image (region labels)

1	1	1	2	2
1	1	1	2	2
1	1	2	2	2
3	1	2	2	2
3	3	2	2	2

Region A contains 9 pixels, region B contains 13 pixels and region C contains 3 pixels. Note that the pixel with value 12 at row 4, col 2 joins region A because it is 4-connected to the pixel with value 13 above it, while the pixels with values 48, 47 and 46 form a separate region.

2. Edge Linking and Boundary Detection

Edge detectors such as Canny give edge pixels, but noise and shadows leave gaps. Edge linking connects them in two ways:

- **Local processing:** an edge pixel (x, y) is linked to a neighbouring edge pixel (x_0, y_0) if $|\nabla f(x, y) - \nabla f(x_0, y_0)| \leq E$ (similar magnitude) and $|\alpha(x, y) - \alpha(x_0, y_0)| \leq A$ (similar direction). Example with E = 15 and A = 10°: a neighbour with magnitude 72 and angle 85° compared with an edge pixel of magnitude 80 and angle 90° gives differences of 8 and 5°, so both conditions hold and they are **linked**. A neighbour with magnitude 40 and angle 20° fails both and is rejected.

- **Global processing:** the Hough transform detects straight field borders (which are common in farmland) even across large gaps, and morphological closing with a small structuring element bridges the remaining short gaps.

The boundary of a segmented region R is extracted as $\partial R = R - (R \ominus B)$, where B is a 3×3 structuring element and $\ominus$ is erosion; that is, the pixels of R that have at least one 4-neighbour outside R. For the example above, region A has 8 boundary pixels (all but the interior pixel at row 2, col 2), and region C has 3 boundary pixels (all its pixels). The ordered boundary can then be described with a chain code.

3. Results

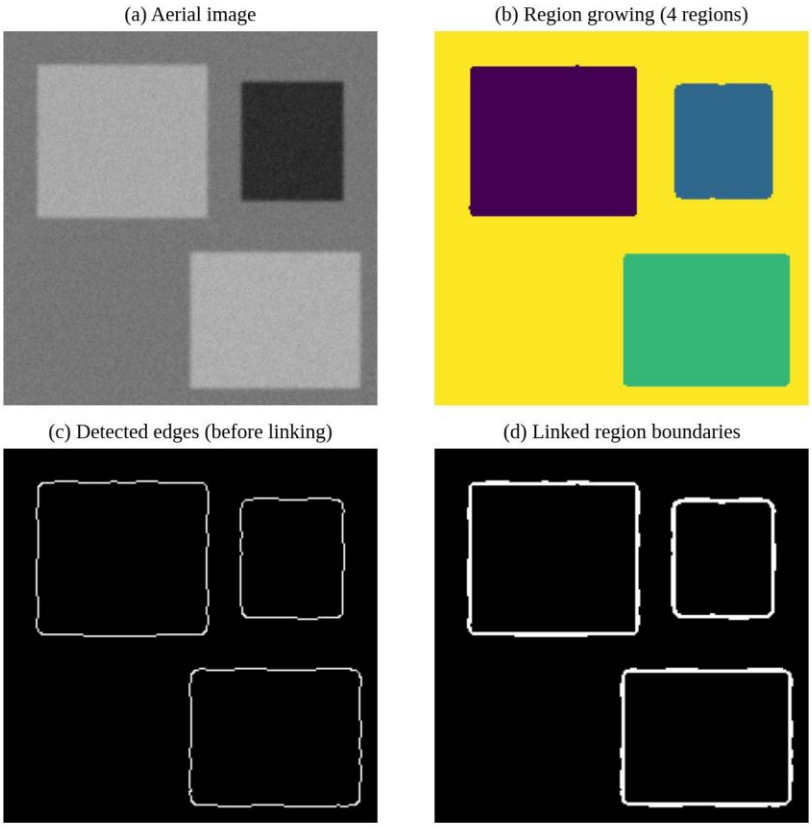


Figure 5: Region growing, edge detection and linked boundaries on an aerial image

Fields A and B have almost the same intensity, so a simple threshold would merge them. Region growing keeps them separate because they are not connected to each other, and the linked boundaries in Figure 5(d) outline each field and the water body as closed contours.

4. How the Techniques Improve Land-Region Identification

- Region growing produces **closed, connected regions** and handles noise by averaging over many pixels.
- Edge detection gives **accurate boundary locations**; linking closes gaps so that every field has a complete outline.
- Combining both is more reliable than either one alone: the edges stop the region from leaking across weak boundaries, and the regions confirm which edges are real field borders.
- Closed boundaries allow area, perimeter and shape of each field to be calculated, which improves crop-area estimation and land-use mapping.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____